



Review article

Diabetic retinopathy pathogenesis and the ameliorating effects of melatonin; involvement of autophagy, inflammation and oxidative stress



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ABSTRACT

Diabetic retinopathy (DR), a microvascular complication of diabetes mellitus (DM), remains as one of the major causes of vision loss worldwide. The release of pro-inflammatory cytokines and the adhesion of leukocytes to retinal capillaries are initial events in DR development. Inflammation, ER stress, oxidative stress and autophagy are major causative factors involved in the pathogenesis of DR. Diabetes associated hyperglycemia leads to mitochondrial electron transport chain dysfunction culminating in a rise in ROS generation. Since mitochondria are the major source of ROS production, oxidative stress induced by mitochondrial dysfunction also contributes to the development of diabetic retinopathy. Autophagy increases in the retina of diabetic patients and is regulated by ER stress, oxidative stress and inflammation-related pathways. Autophagy functions as a double-edged sword in DR. Under mild stress, autophagic activity can lead to cell survival while during severe stress, dysregulated autophagy results in massive cell death and may have a role in initiation and exacerbation of DR. Melatonin and its metabolites play protective roles against inflammation, ER stress and oxidative stress due to their direct free radical scavenger activities and indirect antioxidant activity via the stimulation antioxidant enzymes including glutathione reductase, glutathione peroxidase, superoxide dismutase and catalase. Melatonin also acts as a cell survival agent by modulating autophagy in various cell types and under different conditions through amelioration of oxidative stress, ER stress and inflammation. Herein, we review the possible effects of melatonin on diabetic retinopathy, focusing on its ability to regulate autophagy processes.

1. Introduction

Diabetic retinopathy (DR), a microvascular complication of diabetes mellitus (DM), remains as one of the major causes of vision loss worldwide [1,2]. By 2030, it is expected that the number of patients with DM will have risen to 552 million in comparison with 366 million in 2011 [3]. According to Yau *et al*, over one-third of diabetic patients have signs of DR and the increasing prevalence of DM suggests that many more people will suffer from DR and its complications in the future [1].

There are two types of DR: (a) nonproliferative diabetic retinopathy (NPDR), which is characterized by microaneurysms and possible

intraretinal hemorrhage, hard exudates and cotton wool spots and (b) proliferative diabetic retinopathy (PDR) in which neovascularization of eye fundus and iris besides vitreous hemorrhage are the hallmarks of the disease [4]. Another complication of DR is diabetic macular edema (DME), which is caused by breakdown of outer blood retinal barrier and fluid accumulation [5]. Under hyperglycemic conditions, adhesion of leukocytes to retinal capillaries contributes to endothelial damage and the subsequent increase in vascular permeability results in macular edema and capillary occlusion. As a result of capillary occlusion, retinal ischemia stimulates vascular endothelial growth factor (VEGF) production that is responsible for neovascularization and PDR. Recent studies underline the involvement of inflammation, oxidative stress and

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